

A Pre-feasibility for the Establishment of Asphalt Recycling Plant



2017



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1. Executive Summary

Recycling processes is considered as a part of the production process in many plastic, metal and paper industries, and other industries in order to reduce production costs, conserve natural resources and protect the environment from the residues with negative impacts on communities.

The demand for recycling in the asphalt sector has been increasing in the recent period after the development of alternatives and technical solutions in recycling for economic reasons resulting from the increase of oil prices and therefore the materials used in the asphalt industry. It also helps to maintain the value of natural resources.

The project aims to create asphalt recycling plant and using the recycled asphalt in the production of new asphalt mixes and paving projects that require the presence of asphalt, Ma'an has been selected as the proposed site for the project, given that there are many projects and future plans to improve and roads opening in the south of Jordan.

Table1 : Project Main Indicators

Total Investment cost (JD)	5,050,191
Internal Return Rate	27.96%
Payback Period (Years)	5
Net Present Value (JD)	7,312,155
Job Vacancies	41
Annual Revenue Expectations for 10 years (JD)	76,000,000
Total Operating Cost for 10 years (JD)	56,582,205
Gross Profit for 10 years (JD)	9,224,261

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

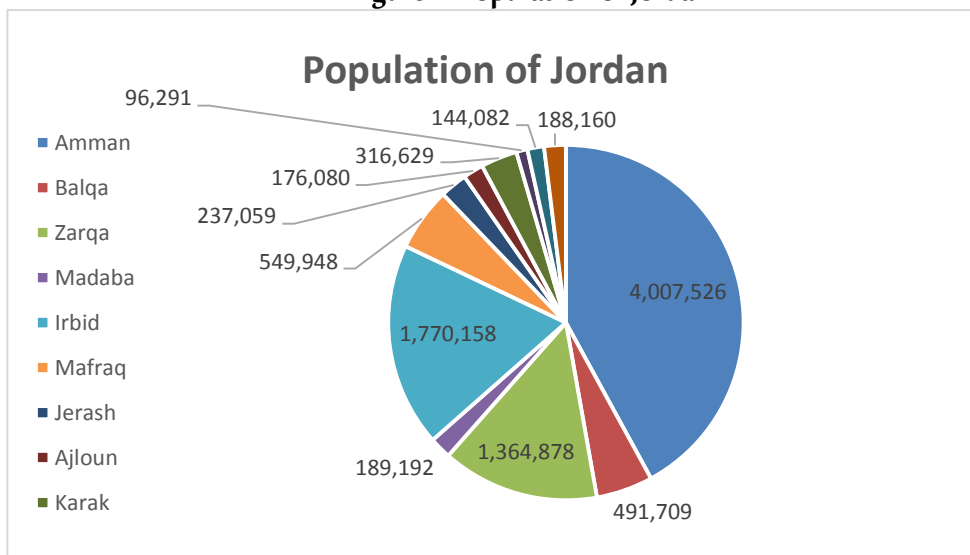
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 1: Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

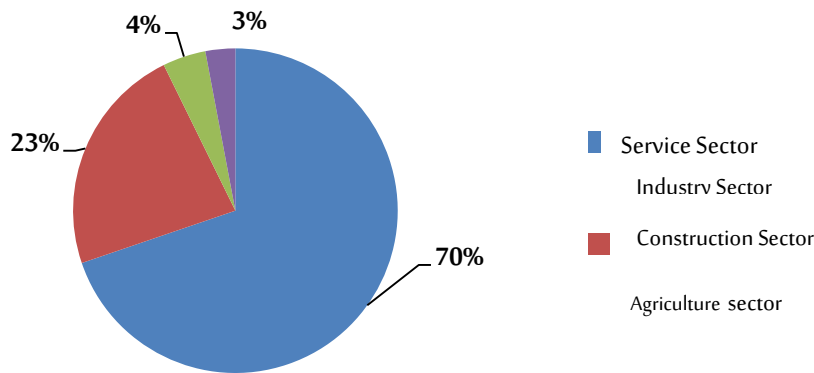
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce. Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Project Description

The company's strategy is based on a clear vision and mission aimed to establish a plant for asphalt recycling in order to preserve the environment and reduce production costs in a way that achieves the highest levels of satisfaction among stakeholders and achieve the best return for its shareholders without losing sight of its role in serving the community and contribution to the national economy development.

The company will purchase and implement the best asphalt recycling technology that would save costs and based on the obligation to protect the environment and streets maintenance.

The asphalt recycling technology for streets and roads contributes in preserving the environment and waste of raw materials, work has already started in this field 30 years ago. This method has proved its worth as there is equipment that increased production quality and improved the quality.

The idea of the project is based on the following economic and marketing reasons and justifications:

- Increasing local, regional and global demand for recycling products such as asphalt, metals and paper.
- The increase of oil and asphalt global prices.
- The high prices of gravel and sands used in paving streets.

Project Objectives

The Objectives of the project are summarized as follows:

- The construction of the asphalt recycling plant is one of the methods that contribute to the preservation of the environment and the waste of raw materials, as work has started in this field 30 years ago. This method has proved its worth as there are equipment that increased production quality and improve the quality.
- Recycling asphalt that was obtained from streets maintenance operations.
- Saving in gravel and sand materials costs used in asphaltting streets by (40-50%).
- The preservation of the environment.
- Maintaining the streets quality and continuous maintenance operations.

The Proposed Service

The main idea of the project and the provided service is the recycling of asphalt mixed used on the streets.

The advantages of the project consist of:

- 1- Saving in gravel and sand materials costs used in asphaltting of streets by (40-50%).
- 2- The preservation of the environment using the (R.A.P) materials.
- 3- Maintaining the streets quality.

Asphalt recycling means the production of new asphalt mixtures consisting of crumbled old asphalt and new debris. The asphalt link consists of the old asphalt link, the new asphalt link and the softening agent of old asphalt link if necessary, and this technique is used to preserve natural resources and save on the allocation of street layers construction and maintenance, and help to protect the environment by recycling.

In addition to many benefits such as: improving the structural capacity of the asphalt with a slight increase in thickness, and fixing layers defects such as creeping and cracking, and the adjustment of the defects of asphalt mixtures and the associated defects of the layers of the streets, it is also considered an economical choice.

Target Segments

The main target for this project is construction companies in Ma'an Governorate in particular, and the contractors in Jordan in general, where the project is not limited to Ma'an Governorate only; But extends to all parts of the Kingdom and serves all construction companies that work in the field of asphalt to reduce the cost and recycling of old asphalt.

Market Size Analysis

In recent years the population of Jordan has increased significantly with the continuation of this pattern. Based on the above, and taking into account the increasing pressure on the transportation network in Jordan, the Ministry of Public Works and the Jordanian Construction Contractors Association as part of its priorities to develop a comprehensive plan for the development of the transportation network in the future. The following table shows some future projects in Jordan:

Table 2 : Future Projects in Jordan

Jordan Streets Future Projects	
Project	Governorate
Asphalt Pavement	Ajloun
Opening and paving the Karak tourist beach	Al-Karak
Streets Paving	Irbid
Paving with hot asphalt mixes in different areas	Amman
Opening and paving streets and sewage systems	Ajloun
Paving with hot asphalt mixes	Al-Mafraq
Opening streets	Al-Balqa
Paving the Jubilee street	Al-Karak
Hot Asphalt Mixture	Jerash
Paving the faculty of technological Engineering Parking lot	Al-Balqa
Hot Asphalt Paving	Al-Balqa
Design and implementation of the internal road - the southern industrial zone	Al-Aqaba
Asphalt mixture	Az Zarqa
Construction and paving the road from A'lyah area to Deban	Maadba
Typical geometric bumps Project	Al-Balqa

In addition to Al-hezam Al-Da'eri project in Irbid, which is 53 km long, the project is still in its first stage, which consists of 17 km and has not yet ended, and many projects to open new roads and maintenance works of the old roads throughout the Kingdom.

The previous table gives a clear idea of the demand for asphalt in the coming years, as Jordan is in a rapid growth phase, which makes the need for a modern road network and new inevitable infrastructure.

Competitors Analysis

There is no factory in Jordan operating in the field of asphalt recycling, and there are several competing factories that manufacture asphalt mixtures, including:

- Abu Shraikh for Roads and Contracting
- Ahmad Yousef Al-Tarawneh and his partner
- The East for Rock Breakers and Public Undertakings
- Haymour for Asphalt Mixtures
- Al-Masar for Contracting
- Al-Awsat for Contracting
- The United for Asphalt Mixtures
- Al-Manhal for Roads and Contracting
- Horizon for rock Breakers and Asphalt Mixtures
- Adel Abu-Sall for Contracting
- Hussien Attyeh for Contracting
- Al-Ja'afar for Contracting

Price Analysis and Pricing Policy

Asphalt mixtures prices range in the Jordanian market between 35-40 JD per ton, this variation and change in price is due to its association with the prices of petroleum products in the market.

The price of the product for this project will be competitive and below the market price, as it consumes fewer raw materials than the new asphalt mixture and therefore prices will be between 26-35 JD per ton.

Market Share

The market share of this project varies as the market depends on the government and private construction projects existing in a given period. The market share varies from year to year. The production capacity of the project will be 200 thousand tons annually.

Projected Revenues Analysis

- The expected revenues for 10 years are as follows:

Table 3 : Revenues of the project during the years 2018-2027

Income sources	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Production capacity percentage	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%
Total Revenues	4000000	4800000	5600000	6400000	7200000	8000000	8800000	9600000	10400000	11200000

The projected annual revenues for the project were calculated based on the analysis of the production capacity of the project as the following table shows:

Table4 Production Capacity

The Analysis of the Production Capacity	
Production Capacity (Tons)	200000
The Required amount for Paving 1 sqm (ton)	0.125
Paving Price (JD / 1 sqm)	10
Cost (JD / 1 sqm)	4

4. Technical Analysis

The Required Human Resources

Table 5 : Required Manpower

Job Title	No.
Indirect Employees	
General Manager	1
Financial Manager	1
Sales & Marketing Manager	1
Secretary	1
Accountants	4
Sales & Marketing Employees	4
Human Resources Manager	1
Human Resources Employees	2
Maintenance Manager	1
Maintenance Employees	2
Total	18
Direct Employees	
Scrape Officer	1
Engineers	2
Production Manager	1
Purchasing & Warehouses Manager	1
Supervisors	2
Workers	10
Production Employees	2
Purchasing & Warehouses Employees	2
Quality Employee	2
Total	10

- **Financial and Administrative Affairs**

- Ensure the proper implementation of financial policies and procedures.
- Record all financial transactions and confirm the correct implementation of the procedures.
- Record and control registers relating to cash management (cash on hand, cash at banks, credit facilities and bank guarantees). Do studies as well to reduce costs.
- Organize and archive all financial documents and constantly update them in an orderly manner.
- Follow-up of all human resources procedures consistently and recruitment, training and salaries, vacations and other.
- Follow-up on the readiness of information technology.

- **Purchasing and Warehouses**

- Check the actual inventory of raw materials and spare parts and make Observations about the levels of differences between the accounting quantities and actual quantities.
- Carry out the purchase and supply in the factory, including a review of procurement and supplier prices, wages and shipping processes requests for delivery, distribution of procurement functions, provide advice in order to ensure alignment of materials and services to the approved specifications.
- Analyze supply requirements, developing resources, recommending the process of analyzing and evaluating suppliers and keep a list of approved suppliers in line with the company's adopted procedures and policies.
- Ensure the preparation of Procurement status reports, provide a summary of progress and submit reports on the prices of materials and industrial trends

- **Production**

- Control the activities of material and production staff, maintenance, safety and quality.
- Examine the project site and ensure adherence to the design specifications and standards of public safety.
- Review and analyze the production plans, preventive maintenance plans, material request orders and reports of quality control and testing.
- Plan and implement the standard operating procedures, developing the key performance indicators and monitoring these indicators.
- Ensure and control the production processes and provide the required human resources. Ensure that key elements are constantly trained and developed in order to increase productivity.
- Follow-up on functions of various units that affect the production and ensure problem-solving between different functions which guarantee the smooth distribution of tasks and be in line with the needs of sales and marketing and factory goals.

- **Maintenance**

- Perform all the maintenance operations at the plant by the maintenance staff.
- The development of preventive and corrective maintenance plans for production lines and ensure compliance with these plans.
- Coordinate with the production and quality control departments regarding the timing and type of maintenance without affecting the target production volume.

Site Analysis

The project will be established in Ma'an Governorate, specifically in Al Rawdah Industrial Area, where it is considered one of the development areas in Jordan, which is suitable for the establishment of factories.

Technical Requirements Analysis

Land and construction works

The project will be constructed on land with an area of 12,000 sqm. The following table shows the required land:

Table6 : The required land

Land	Area (sqm)
The required land	12,000

The required construction works for the project is represented as follows:

Table 7 : The Required Construction Works

Construction Works	Area (sqm)
Hanger	1,250
Buildings	750
Paving the ground	150

Machinery and Equipment

The following table shows the machinery and equipment needed for the project:

Table 8 : The Required Machinery and Equipment

Machinery and Equipment	Quantity
Asphalt Mixer	1
Scraper with 2-meter width	2
Tipper Truck	3
Electrical Generator	1
Balance	1
Quality Laboratory	1
Street Sweeper	2
Loader	2
Road Roller	3

Furniture

The project needs furniture for the administrative offices.

Vehicles and Transportation

The project needs Vehicles as following:

Table 9 : The Required Vehicles

Vehicles	Quantity
Pickup Trucks	2

Raw Materials

Asphalt mixture consists of several components and these components are added in varying proportions depending on the nature of the place to be paved. These materials are:

Table 10 : Raw Material

The Used Raw Materials
Limestone
Gravel and Sand
Fuel
Liquid Asphalt

Table11 : The required raw materials to produce ton of recycled asphalt

Material	Quantity (KG)	Percentage	Average Price (JD / Ton)	Producing one ton cost
Recycled Material	450	%45	-	
Gravel and Sand	490	%49	10	4.9
Liquid Asphalt	50	%5	340	17
Fuel	10	%1	320	3.2

5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.05% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 12: General Assumption

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 13: Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 14: Annual Growth Rates Assumptions

	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 15: Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	40.00%
Utilities Expenses from Total Revenues	12.00%
Maintenance Expenses from Total Revenues	2.00%
Depreciation Expenses from Total Revenues	10.00%
Insurance Expenses from Total Assets Value	0.50%
Marketing Expenses from Total Revenues	0.50%

Table 16: Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	5%
(2) Income Tax Deduction	0%
Income Tax after Deduction	5%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Table 17: Risk Premium

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premium	10.12%
Income Tax Rate	5.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 18: Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%5
Bets	1.00
Equity	2,020,076
Loans Value	3,030,114

Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	3,030,114
Loans Percentage	%60
Equity	
Cost of Equity	%16.6
Equity Value	2,020,076
Equity Percentage	%40
Gross	
Project Value	5,050,191
Weighted Average Cost of Capital	%100

Capital Expenditures

The estimated capital costs of the project are 5,050,191 JD and cover the cost of land, construction, machinery and equipment, furniture and furnishings, vehicles and transportation, pre-operating expenses and initial working capital. These costs are shown in the following tables:

Table 19: Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost
Required Land	5	12,000	60,000
Gross Cost			60,000

Table 20: Construction Work Capital Expenditure

Construction Works	Cost (JOD per Squared	Area	Gross Cost
Hanger	100	1,250	125,000
Construction Works	250	750	187,500
Paving	12	150	1,800
Other			31,430
Gross Cost			345,730

Table 21: Machinery and equipment Capital Expenditure

Machinery and equipment	Quantity	Unit cost	Gross Cost
Asphalt Mixer	1	1,000,000	1,000,000
Scraper with 2 meter width	2	180,000	360,000
Tipper Truck	3	400,000	1,200,000
Electrical Generator	1	50,000	50,000
Balance	1	30,000	30,000
Quality Laboratory	1	65,000	65,000
Street Sweeper	2	75,000	150,000
Loader	2	60,000	120,000
Road Roller	3	35,000	105,000
Transportation and installation			150,000
Gross Total (JD)			3,230,000

Table 22 : Furnitutre Capital Expenditure

Furnitutre	Quantity	Unit cost	Gross Cost /Jordan Dinar
Office Furnitutre	-	-	18,000
Total			18,000

Table 23: Pre-operating expenses

Pre-operating expenses	Gross Cost
Governmental fees	1,500
Transportation	1,500
Marketing	1,500
Training	1,500
Total	6,000

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 24: Initial working capital

Initial working capital	Number of months	Percentage	Cost (JD)
Operating expenses	3	%25	767,800
Administrative and general expenses	3	%25	15,752
Marketing expenses	3	%25	5,000
Indirect salaries	3	%25	72,600
Gross Total			861,152

Table 25: Summary of expenses

Summary of capital expenditures	Cost (JD)	Contingency	Cost (JD)	Percentage
Land	60,000	10%	66,000	%1.3
Construction Works	345,730	10%	380,303	%7.5
Machinery and Equipment	3,230,000	10%	3,553,000	%70.4
Furniture	18,000	10%	19,800	%0.4
Vehicles and Transportation	70,200	10%	77,220	%1.5
Pre-operating expenses	6,000	10%	6,600	%0.1
Initial working capital	861,152	10%	947,268	%18.8
Total (JD)	4,591,082		5,050,191	100%

Sources of funding

The financing structure required for the project consists of loans and equity. The investment cost of this project was estimated at JD 5,050,191. This amount is determined through fixed assets, which include land, construction, machinery and equipment, furniture and furnishings, vehicles and transport vehicles, pre-operating expenses and initial working capital. (60%) of the project will be financed through loans (3,030,114JD) and the rest (40%) through self financing (equity). The following table shows the general structure of the required financing:

Table 26: Sources and use of finance

Sources of funding	amount	Percentage
Equity (self-financing)	2,020,076	%40
Loans	3,030,114	%60
Gross Total	5,050,191	%100
Use of funding	amount	Percentage
Capital expenditure	4,096,323	%81.11
Pre-operating expenses	6,600	%0.13
Working capital	947,268	%18.76
Gross Total	5,050,191	%100.00

Operating Expenses

The following table shows the expected operating expenses of the project:

Table 27: Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw material expenses	1,600,000	1,920,000	2,240,000	2,560,000	2,880,000	3,200,000	3,520,000	3,840,000	4,160,000	4,480,000
Service expenses	480,000	576,000	672,000	768,000	864,000	960,000	1,056,000	1,152,000	1,248,000	1,344,000
Maintenance expenses	80,000	96,000	112,000	128,000	144,000	160,000	176,000	192,000	208,000	224,000
Consumables	400,000	480,000	560,000	640,000	720,000	800,000	880,000	960,000	1,040,000	1,120,000
Direct salaries	232,000	241,512	251,414	261,722	272,453	283,623	295,252	307,357	319,959	333,077
Other expenses	279,200	331,351	383,541	435,772	488,045	540,362	592,725	645,136	697,596	750,108
Gross Total	3,071,200	3,644,863	4,218,955	4,793,494	5,368,498	5,943,985	6,519,977	7,096,493	7,673,554	8,251,185

General and Administrative Expenses

The following table shows the expected administrative and general expenses of the project:

Table 28: administrative and general expenses

description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Communications expenses	4,800	4,944	5,092	5,245	5,402	5,565	5,731	5,903	6,080	6,263
Hospitality expenses	2,400	2,472	2,546	2,623	2,701	2,782	2,866	2,952	3,040	3,131
Stationery and prints	2,400	2,472	2,546	2,623	2,701	2,782	2,866	2,952	3,040	3,131
Transportation and movements	4,800	4,944	5,092	5,245	5,402	5,565	5,731	5,903	6,080	6,263
Rent expenses	20,000	20,600	21,218	21,855	22,510	23,185	23,881	24,597	25,335	26,095
Training expenses	2,400	2,472	2,546	2,623	2,701	2,782	2,866	2,952	3,040	3,131
Insurance expenses	20,482	21,096	21,729	22,381	23,052	23,744	24,456	25,190	25,945	26,724
Other expenses	5,728	5,900	6,077	6,259	6,447	6,641	6,840	7,045	7,256	7,474
Gross Total	63,010	64,900	66,847	68,852	70,918	73,046	75,237	77,494	79,819	82,213

Marketing Expenses

Table 29: Marketing Expenses

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	20,000	24,000	28,000	32,000	36,000	40,000	44,000	48,000	52,000	56,000

Human Resources

The salaries and monthly wages of each employee were estimated to be multiplied by 16 months of the year for the purpose of calculating the increase in salaries and annual wages due to social security, health insurance, etc. to the employees of the administration. The following table shows the expected annual salaries and wages. The annual rate of increase in wages and salaries is taken into account:

Table 30: Expected monthly salaries for the years (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
General Manager	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
Financial Manager	2,500	2,603	2,709	2,820	2,936	3,056	3,182	3,312	3,448	3,589
Sales & Marketing Manager	2,000	2,082	2,167	2,256	2,349	2,445	2,545	2,650	2,758	2,871
Secretary	800	833	867	902	939	978	1,018	1,060	1,103	1,149
Accountants	600	625	650	677	705	734	764	795	827	861
Sales & Marketing Employees	500	521	542	564	587	611	636	662	690	718
Human Resources Manager	1,750	1,822	1,896	1,974	2,055	2,139	2,227	2,318	2,413	2,512
Human Resources Employees	500	521	542	564	587	611	636	662	690	718
Maintenance Manager	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Maintenance Employees	350	364	379	395	411	428	445	464	483	502
Direct Employees										
Scrape Officer	2,500	2,603	2,709	2,820	2,936	3,056	3,182	3,312	3,448	3,589
Engineers	700	729	759	790	822	856	891	927	965	1,005

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Production Manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Purchasing & Warehouses Manager	1,400	1,457	1,517	1,579	1,644	1,712	1,782	1,855	1,931	2,010
Supervisors	750	781	813	846	881	917	954	994	1,034	1,077
Workers	300	312	325	338	352	367	382	397	414	431
Production Employees	500	521	542	564	587	611	636	662	690	718
Purchasing & Warehouses Employees	500	521	542	564	587	611	636	662	690	718
Quality Employee	600	625	650	677	705	734	764	795	827	861

Table 31: Expected yearly salaries for the years (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
General Manager	64,000	66,624	69,356	72,199	75,159	78,241	81,449	84,788	88,264	91,883
Financial Manager	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Sales & Marketing Manager	32,000	33,312	34,678	36,100	37,580	39,120	40,724	42,394	44,132	45,942
Secretary	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Accountants	38,400	39,974	41,613	43,319	45,096	46,945	48,869	50,873	52,959	55,130
Sales & Marketing Employees	32,000	33,312	34,678	36,100	37,580	39,120	40,724	42,394	44,132	45,942
Human Resources Manager	28,000	29,148	30,343	31,587	32,882	34,230	35,634	37,095	38,616	40,199
Human Resources Employees	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Maintenance Manager	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Maintenance Employees	11,200	11,659	12,137	12,635	13,153	13,692	14,254	14,838	15,446	16,080
Total indirect salaries	290,400	302,306	314,701	327,604	341,035	355,018	369,574	384,726	400,500	416,920
Direct Employees										
Scrape Officer	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Engineers	22,400	23,318	24,274	25,270	26,306	27,384	28,507	29,676	30,893	32,159
Production Manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Purchasing & Warehouses Manager	22,400	23,318	24,274	25,270	26,306	27,384	28,507	29,676	30,893	32,159
Supervisors	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Workers	48,000	49,968	52,017	54,149	56,369	58,681	61,087	63,591	66,198	68,912
Production Employees	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Purchasing & Warehouses Employees	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Quality Employee	19,200	19,987	20,807	21,660	22,548	23,472	24,435	25,436	26,479	27,565
Total direct salaries	232,000	241,512	251,414	261,722	272,453	283,623	295,252	307,357	319,959	333,077
المجموع	522,400	543,818	566,115	589,326	613,488	638,641	664,825	692,083	720,459	749,997

Depreciations

:The following tables show all equipment costs, expenses and depreciation rates

Table 32: Capex and Depreciation Expenses

Costs of equipment, capital expenditures and annual depreciation	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	66,000	%0.0	%0.0
Construction Works	380,303	%5.0	%0.0
Machinaries	3,553,000	%10.0	%2.0
Furniture	19,800	%10.0	%5.0
Vehicles	77,220	%10.0	%0.0
	4,096,323		

Table 33: Depreciation and annual increases on construction work

Summary of capital costs, additions and annual depreciation	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total fixed assets	4,096,3	4,168,3	4,241,8	4,316,9	4,393,4	4,471,5	4,551,3	4,632,6	4,715,6	4,800,4
Total consumption expenses	384,017	391,222	398,574	406,076	413,732	421,544	429,516	437,651	445,953	454,425
Total accumulated depreciation	384,017	775,239	1,173,8	1,579,8	1,993,6	2,415,1	2,844,6	3,282,3	3,728,2	4,182,7
Total additions	0	72,050	73,521	75,022	76,555	78,121	79,720	81,352	83,019	84,721
Total net book value	3,712,3	3,393,1	3,068,0	2,737,0	2,399,8	2,056,4	1,706,6	1,350,3	987,396	617,692

Loans

The following table represents the annual loan amount and the annual payment method with the details of the remaining amounts of loans for each year.

Table 34: Loan Details

Loan Value	3,030,114
Annual Interest Rate	9.48%
Loans Period	5
Grace Period	0
Loan Starts at year:	2018
Annual Payment	788,746
Number of Payments	5

Year	Annual Payment	Interest	Equity capital	Loan Remaining Value
2018				3,030,114
2019	788,746	287,255	501,491	2,528,623
2020	788,746	239,714	549,032	1,979,591
2021	788,746	187,665	601,081	1,378,510
2022	788,746	130,683	658,063	720,447
2023	788,746	68,298	720,447	0

Income Statement

Table 35: Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	4,000,00	4,800,00	5,600,00	6,400,00	7,200,00	8,000,00	8,800,00	9,600,00	10,400,00	11,200,00
Gross Operating Revenues	4,000,00	4,800,00	5,600,00	6,400,00	7,200,00	8,000,00	8,800,00	9,600,00	10,400,00	11,200,00
Operating Expenses	3,071,20	3,644,86	4,218,95	4,793,49	5,368,49	5,943,98	6,519,97	7,096,49	7,673,554	8,251,185
Gross Operating Profit	928,800	1,155,13	1,381,04	1,606,50	1,831,50	2,056,01	2,280,02	2,503,50	2,726,446	2,948,815
<i>Gross Profit Percentage</i>	%23	%24	%25	%25	%25	%26	%26	%26	%26	%26
Salaries and Benefits (Indirect Staff)	-290,400	-302,306	-314,701	-327,604	-341,035	-355,018	-369,574	-384,726	-400,500	-416,920
General and Administraive Expenses	-63,010	-64,900	-66,847	-68,852	-70,918	-73,046	-75,237	-77,494	-79,819	-82,213
Markeing Expenses	-20,000	-24,000	-28,000	-32,000	-36,000	-40,000	-44,000	-48,000	-52,000	-56,000
Pre Operating Expenses	-6,600									
Gross Indirect Expenses	-380,010	-391,206	-409,548	-428,456	-447,954	-468,064	-488,811	-510,220	-532,319	-555,134
Income before Interest, Depreciation	548,790	763,930	971,497	1,178,05	1,383,54	1,587,95	1,791,21	1,993,28	2,194,127	2,393,681
Fixed Assets Depreciations	-384,017	-391,222	-398,574	-406,076	-413,732	-421,544	-429,516	-437,651	-445,953	-454,425
Income before Tax and Interests	164,773	372,708	572,922	771,973	969,817	1,166,40	1,361,69	1,555,63	1,748,174	1,939,256
Bank Interests	-287,255	-239,714	-187,665	-130,683	-68,298	-0	-0	-0	-0	-0
Income Before Tax	-122,482	132,995	385,257	641,290	901,518	1,166,40	1,361,69	1,555,63	1,748,174	1,939,256
Income Tax	0	-526	-19,263	-32,065	-45,076	-58,320	-68,085	-77,782	-87,409	-96,963

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Profit	-122,482	132,469	365,994	609,226	856,442	1,108,08	1,293,61	1,477,85	1,660,765	1,842,293
<i>Net Profit Percentage</i>	<i>%3-</i>	<i>%3</i>	<i>%7</i>	<i>%10</i>	<i>%12</i>	<i>%14</i>	<i>%15</i>	<i>%15</i>	<i>%16</i>	<i>%16</i>
Compulsory Reserves	0	-13,247	-36,599	-60,923	-85,644	-110,809	-129,361	-147,785	-166,076	-184,229
Retained Earnings	-122,482	-3,260	326,135	874,438	1,645,23	2,642,51	3,806,76	5,136,83	6,631,522	8,289,586

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 36: Expected Cashflows Statement

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating										
Net Profit	-122,482	132,46	365,99	609,226	856,442	1,108,08	1,293,61	1,477,85	1,660,76	1,842,293
Non-monetary adjustments										
Bank Interests	287,255	239,71	187,66	130,683	68,298	0	0	0	0	0
Depreciation	384,017	391,22	398,57	406,076	413,732	421,544	429,516	437,651	445,953	454,425
Total Operating Cashflows before Additions to Working Capital	548,790	763,40	952,23	1,145,98	1,338,47	1,529,63	1,723,12	1,915,50	2,106,71	2,296,719
Inventory (Increase/Decrease)	-133,333	-26,667	-26,667	-26,667	-26,667	-26,667	-26,667	-26,667	-26,667	-26,667
Accounts Receivables	-333,333	-66,667	-66,667	-66,667	-66,667	-66,667	-66,667	-66,667	-66,667	-66,667
Accounts Payables	367,800	63,416	63,523	63,635	63,751	63,872	63,998	64,129	64,265	64,408

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Working Capital (Increase/Decrease)	-98,867	-29,918	-29,810	-29,699	-29,582	-29,461	-29,335	-29,204	-29,068	-28,926
Net Cashflows from Operating Activities	449,924	733,48	922,42	1,116,28	1,308,89	1,500,16	1,693,79	1,886,30	2,077,65	2,267,793
Cashflows from Investments										
Fixed Assets (Procurement)	4,096,32	-72,050	-73,521	-75,022	-76,555	-78,121	-79,720	-81,352	-83,019	-84,721
Net Cashflows from Investments Activities	4,096,32 -3	-72,050	-73,521	-75,022	-76,555	-78,121	-79,720	-81,352	-83,019	-84,721
Cashflows from Financing Activities										
Capital	2,020,07									
Loan Amortization	-501,491	549,03	601,08	-658,063	-720,447	0	0	0	0	0
Bank Interest Rate	-287,255	239,71	187,66	-130,683	-68,298	-0	-0	-0	-0	-0
Loans	3,030,11	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	4,261,44 5	788,74 -6	788,74 -6	-788,746	-788,746	0	0	0	0	0
Net increase (decrease) in cash	615,045	127,30	60,157	252,518	443,589	1,422,04	1,614,07	1,804,94	1,994,63	2,183,072
Cashflows at the Beginning of Period	0	615,04	487,73	547,894	800,412	1,244,00	2,666,04	4,280,12	6,085,07	8,079,703
Cashflows at the end of Period	615,045	487,73	547,89	800,412	1,244,00	2,666,04	4,280,12	6,085,07	8,079,70	10,262,77

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 37: Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	615,045	487,737	547,894	800,412	1,244,001	2,666,049	4,280,122	6,085,071	8,079,703	10,262,775
Inventory	133,333	160,000	186,667	213,333	240,000	266,667	293,333	320,000	346,667	373,333
Accounts receivable	333,333	400,000	466,667	533,333	600,000	666,667	733,333	800,000	866,667	933,333
Total assets traded	1,081,712	1,047,737	1,201,227	1,547,079	2,084,001	3,599,383	5,306,789	7,205,071	9,293,036	11,569,441
Non-current assets										
Fixed assets (net)	3,712,306	3,393,134	3,068,080	2,737,026	2,399,850	2,056,426	1,706,630	1,350,330	987,396	617,692
Total non-current assets	3,712,306	3,393,134	3,068,080	2,737,026	2,399,850	2,056,426	1,706,630	1,350,330	987,396	617,692
Total assets	4,794,018	4,440,870	4,269,307	4,284,105	4,483,850	5,655,809	7,013,419	8,555,402	10,280,432	12,187,133
Liabilities	-	-	-	-	-					
Accounts Payables	367,800	431,216	494,739	558,374	622,124	685,996	749,994	814,123	878,389	942,796
Remaining amount of Loan	549,032	601,081	658,063	720,447	0-	0-	0-	0-	0-	0
Total current liabilities	916,832	1,032,296	1,152,802	1,278,821	622,124	685,996	749,994	814,123	878,389	942,796

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Non-current liabilities										
long term loans	1,979,591	1,378,510	720,447	0	0	0	0	0	0	0
Total long - term liabilities	1,979,591	1,378,510	720,447	0	0	0	0	0	0	0
Total liabilities	2,896,423	2,410,807	1,873,249	1,278,821	622,124	685,996	749,994	814,123	878,389	942,796
Property rights										
Members Contributions	2,020,076	2,020,076	2,020,076	2,020,076	2,020,076	2,020,076	2,020,076	2,020,076	2,020,076	2,020,076
Compulsory reserve	0	13,247	49,846	110,769	196,413	307,222	436,583	584,368	750,445	934,674
Profit (loss)	122,482-	3,260-	326,135	874,438	1,645,237	2,642,515	3,806,765	5,136,834	6,631,522	8,289,586
Total equity	1,897,595	2,030,064	2,396,058	3,005,284	3,861,726	4,969,813	6,263,424	7,741,278	9,402,043	11,244,337
Total liabilities and equity	4,794,018	4,440,870	4,269,307	4,284,105	4,483,850	5,655,809	7,013,419	8,555,402	10,280,432	12,187,133

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 38: Free Net Cash flows Table

Net Cashflows	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	-5,050,191	456,524	661,437	848,903	1,041,26	1,232,33	1,422,048	1,614,07	1,804,94	1,994,63	2,183,07	28,198,05
Discount Factor	1.00	0.71	0.63	0.57	0.51	0.45	0.40	0.36	0.32	0.29	0.26	0.26
Net Present Value for Cash Flows	-5,050,191	324,496	419,582	480,582	526,081	555,651	572,229	579,643	578,475	570,511	557,252	7,197,845

Table below (38) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 39: payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-5,050,191	456,524	661,437	848,903	1,041,26	1,232,33	1,422,04	1,614,07	1,804,94	1,994,63	2,183,07
Project Value	5,050,191	4,593,66	3,932,23	3,083,32	2,042,06	809,728	-612,320	2,226,39	4,031,34	6,025,97	8,209,04
Payback Period	5	1	1	1	1	1	0	0	0	0	0

Table 40: Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%12.05
Net Present Value for Cashflows	7,312,155
Payback Period	5
Internal Rate of Return	%27.96

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 7,312,155 Jordan Dinars considering that the project provides 41 Job Opportunities for the governorate residents.

Table 41: Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%27.96	5	12.05%
Revenues declined by 10%	%25.26	6	12.05%
Operating Expenses Increased by 10%	%19.56	7	12.05%